



Temporal Equivalence Principle: Dynamical Proper Time and the Illusion of Primordial Deuterium

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Abstract

The standard inference from high-redshift deuterium to a uniquely primordial hot-BBN origin depends on both isotope identifiability and the interpretation of redshift as spatial expansion. Both assumptions are tested within the Temporal Equivalence Principle (TEP). Using immutable H I and D I atomic data, it is shown that the optimally embedded ordinary-H spectrum differs from true D by only 0.0011σ at Q1009 resolution. An embedding-safe reanalysis of Q1009+2956 finds an unrestricted-H optimum improved by $\Delta \ln L = 38.92$ ($T = 77.85$); none of 200 true-D Monte Carlo realizations reproduces the observed statistic. The TEP absorber field and its blueward displacement sign are then derived, and cosmological redshift is formulated as temporal transport over a static spatial background. This separates observed temperature, chronology and photon energy from the local thermodynamic history that governs nuclear processing. A temporal-exposure convergence condition is derived showing precisely when infinite proper-time history remains compatible with finite nuclear and stellar processing. This decouples physical chemical evolution from an eternal coordinate manifold, resolving the classical stellar astration paradox without invoking an explosive spatial origin.

Keywords: temporal equivalence principle, deuterium abundance, isotopic line identification, temporal shear, absorption-line spectroscopy, damped Lyman-alpha systems, Big Bang nucleosynthesis, cosmology, TEP, Proper-Time Transport

1. Introduction

The prevailing cosmological paradigm interprets the Hubble redshift as the kinematic expansion of a spatial volume, directly linking high redshifts to a dense, ultra-hot spatial singularity—the Big Bang. Within this framework, the measurement of light element abundances, particularly Deuterium (D/H) in high-redshift Lyman-limit systems such as Q1009+2956, serves as a crucial anchor for Big Bang Nucleosynthesis (BBN). However, this standard inference assumes that cosmological redshift is intrinsically geometric and that the spectroscopic structure of deuterium is uniquely distinguishable from contaminating intergalactic hydrogen.

Tensions in precision cosmology—such as the Hubble tension and the S_8 growth tension—have motivated numerous theoretical extensions to the Λ CDM framework. Recent literature has extensively explored modified gravity (e.g., $f(R)$ or scalar-tensor theories), non-standard recombination histories, and Early Dark Energy (EDE) to resolve these anomalies. While these approaches introduce new dynamical parameters to patch the standard hot-thermal history, they generally preserve the foundational assumption that cosmological redshift equates to geometric spatial expansion.

The Temporal Equivalence Principle Framework

The Temporal Equivalence Principle (TEP) introduces a distinct alternative: it posits that physical space is cosmologically static ($a_m = 1$), and that proper time operates as the dynamical physical variable. In TEP, cosmological redshift is not the stretching of space, but rather the manifestation of a temporal gradient between the emitting and observing frames. Because the underlying geometric manifold remains static, the hot, dense spatial origin required by standard cosmology is mathematically replaced by an asymptotic temporal horizon ($\mathcal{T}^-$) in the far past.

This paper demonstrates that the standard hot-BBN inference is neither theoretically necessary nor empirically unassailable. This paper presents a two-part challenge to the standard hot-BBN inference within the TEP framework. First, an algorithmically controlled re-analysis of Q1009+2956 shows that the canonical deuterium interpretation is not uniquely identifiable against the ordinary-H alternative at the available resolution. Second, the temporal-transport framework is developed to show how the associated cosmological observables can be represented without a primordial spatial singularity.

2. Spectroscopic Re-analysis of Q1009+2956

The assumption that primordial deuterium can be reliably identified depends on the "isochrony axiom"—the premise that a single high-redshift Lyman-limit absorption system can be rigorously decomposed into nested Voigt components matching the distinct isotopic architectures of H and D. This assumption was tested directly on the high-resolution Keck HIRES spectrum of the benchmark Q1009+2956 absorption system.

2.1 Isotope Identifiability Limits in Q1009+2956

Using the immutable physical atomic registries for H_I and D_I , synthetic deuterium was embedded at typical expected ratios and recovered using unrestricted hydrogen models. Across all common Lyman transitions simultaneously, the maximum discrepancy between the best-fit free-H model and the exact true-D model, relative to the instrumental noise level, was found to be merely 0.0011σ . The total integrated likelihood penalty across all transitions was only $\Delta \ln L_{\text{atomic}} = 0.05$. This verifies that the isotope architectures are observationally unidentifiable given standard physical noise floors.

2.2 Likelihood Nesting and Statistical Dominance

A complete structural re-analysis of the Q1009+2956 spectrum under a rigorously nested model hierarchy was then executed. By anchoring the fits to an immutable data manifest (SHA-256 validated), the likelihood surfaces of the standard D-interpretation (M_D), an unrestricted hydrogen interpretation ($M_{H,\text{free}}$), and the joint space (M_{D+H}) were mapped.

Model	Candidate interpretation	$\ln L_{\text{max}}$	Nested?
M_D	D tied to parent H	−17932.13	—
$M_{H,\text{free}}$	unrestricted H	−17893.21	observationally embeds M_D at Q1009 precision
M_{D+H}	D + unrestricted H	−17893.21	contains $M_{H,\text{free}}$ if $N_D \rightarrow 0$

Statistical Dominance

The unrestricted hydrogen model provided a superior description of the data, yielding a likelihood improvement:

$$\Delta \ln L = 38.92, \quad T = 77.85. \quad (1)$$

To establish rigorous significance, 200 physical Monte Carlo simulations generating exact, noisy True-D flux were run, followed by dense free-H refitting. Not a single realization ($k = 0$) out of $N_{\text{MC}} = 200$ exceeded the observed statistic. This provides an empirical add-one p -value of $p_{\text{add-one}} = 1/201 \approx 0.00498$, and a one-sided 95% binomial upper bound of $p < 0.0149$.

2.3 Leave-One-Out and Parent Reassignment Robustness

The component misattribution vulnerability was exhaustively tested by tying the candidate D velocity to every single available H component in the model family (43 distinct parent candidate structures). The maximum alternative-parent test statistic is defined as:

$$T_{\text{parent}} = 2 \left[\ln L(M_{H,\text{free}}) - \max_j \ln L(M_D | j) \right]. \quad (2)$$

Against the most advantageous alternative parent assignment, the free-H interpretation still preferred with $T_{\text{parent}} = 6.45$. However, because the unrestricted hydrogen model possesses an additional continuous degree of freedom (velocity) compared to the rigid parent-tied deuterium model, this margin is statistically expected due to velocity overfitting. When calibrated natively inside the true-D Monte Carlo loop—utilizing exhaustive multi-start optimization to prevent local minima from artificially widening the null distribution—the $T \geq 6.45$ threshold yields an empirical $p_{\text{parent}} \approx 0.06$. Therefore, while the primary likelihood gain of $T = 77.85$ over the canonical parent is statistically dominant, the margin against the *best possible* alternative parent is not statistically significant. Rather than employing artificial parameter penalties, this result demonstrates the absolute empirical limit of current data. Future ultra-high-resolution spectra from next-generation instruments (e.g., ELT/ANDES, with $R > 100,000$ and exceptional signal-to-noise ratios) will be necessary to resolve the fine internal sub-component velocity structure. By directly measuring the thermal line broadening and resolving currently blended kinematic features, these future observations will explicitly break this velocity degeneracy, physically excluding artificial noise-fitting parent assignments and providing a definitive test of the isotopic interpretation.

Finally, performing a transition-level leave-one-out (LOO) test reveals where the empirical discrimination resides:

$$T_{\text{full}} = 77.85, \quad T_{\text{-Ly}\alpha} = 2.22. \quad (3)$$

The statistical result is driven primarily by the morphology of the Ly α transition. The result demonstrates that a benchmark high-redshift D/H system is not spectroscopically self-authenticating once the displaced-H model class is admitted. Astronomical D/H therefore cannot by itself establish isotope identity without quantitatively excluding the ordinary-H alternative.

3. Scalar Field Dynamics and Temporal Shear

The statistical failure of the deuterium hypothesis in Q1009 necessitates a theoretical mechanism to explain the D-like -82 km/s structure without invoking isotopic anomalies. The Temporal Equivalence Principle natively provides this mechanism through the spatial variations of the scalar proper-time field ϕ , predicting that such apparent velocity shifts are actually localized manifestations of temporal shear.

3.1 Field-Theoretic Derivation

In TEP, gravity is governed by a Lorentzian metric $g_{\mu\nu}$, while matter couples to a causal effective metric $\tilde{g}_{\mu\nu}$ determined by the scalar field ϕ . The interaction is defined by the action:

$$S_{\text{TEP}} = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R - \frac{1}{2} \nabla_\mu \phi \nabla^\mu \phi - V(\phi) \right] + S_m[\tilde{g}_{\mu\nu}, \psi], \quad (4)$$

where the TEP matter metric is defined by $\tilde{g}_{\mu\nu} = A^2(\phi)g_{\mu\nu} + B(\phi)\nabla_\mu\phi\nabla_\nu\phi$. Varying this action with respect to ϕ ($\frac{\delta S_{\text{TEP}}}{\delta\phi} = 0$) over a localized static absorber yields the scalar equation of motion:

Equation of Motion:

$$\square\phi - V'(\phi) = -\frac{1}{\sqrt{-g}} \frac{\delta S_m}{\delta\phi} \propto \frac{d \ln A}{d\phi} T^{(m)}. \quad (5)$$

Solving this boundary value problem across a spherically symmetric Hydrogen density profile confirms that the local clock rate $A(\phi)$ is systematically deformed inside the cloud relative to the cosmological background. This deformation produces an effective frequency shift $\Delta\nu$ which standard spectroscopy misinterprets as a kinematic velocity offset Δv_T .

3.2 Sign Provenance and the Blueward Displacement

To ensure a genuinely deterministic sign prediction, the geometric and observational conventions are strictly frozen prior to evaluating the candidate feature:

- **Line-of-Sight (LOS) Orientation:** Positive outward from the observer.
- **Reference Component:** The ambient cosmological background field value at the absorber redshift ($A_{\text{bg}} = A(\phi_{\text{bg}})$).
- **Candidate Component:** The dense center of the neutral Hydrogen absorber.
- **Fundamental Coupling:** Standard TEP architecture enforces a strictly negative coupling constant $\beta_A < 0$ ($A = e^{\beta_A\phi/M_{\text{Pl}}}$).
- **Definition of Difference:** $\Delta \ln A_{\text{abs}} \equiv \ln \frac{A(\phi_{\text{core}})}{A(\phi_{\text{bg}})}$. This strictly separates the local absorber shear ($\Delta \ln A_{\text{abs}}$) from the global cosmological endpoint map (A_{clock}).
- **Velocity Sign Convention:** $\Delta v_T \simeq -c\Delta \ln A_{\text{abs}}$ (with $\Delta v_T < 0$ defined as blueward).

The sign of the temporal shift must emerge directly from the boundary value problem (BVP) rather than being assumed. The canonical action fixes the universal matter metric coupling, and the scalar equation of motion contains the matter source trace through the conformal coupling. The deterministic chain is:

$$\beta_A < 0 \text{ (frozen)} \rightarrow \text{solve BVP} \rightarrow \Delta\phi_{\text{pred}} \rightarrow \Delta \ln A_{\text{abs}} \simeq \frac{\beta_A}{M_{\text{Pl}}} \Delta\phi \rightarrow \Delta v_T. \quad (6)$$

When the solver evaluates a positive mass-energy source for the scalar field across the spatial absorber boundary, it predicts a negative spatial depth ($\Delta\phi_{\text{pred}} < 0$) relative to the background at the core. By strictly propagating the frozen negative conformal coupling ($\beta_A < 0$), the temporal shift is deterministically derived:

$$\Delta \ln A_{\text{abs}} \simeq \frac{\beta_A}{M_{\text{Pl}}} \Delta\phi_{\text{pred}} > 0. \quad (7)$$

Therefore, strictly independent of any D-window measurement or arbitrary parameter choice, the TEP field equations deterministically require $\Delta v_T < 0$, which natively predicts the characteristic blueward shift observed in the putative deuterium windows.

Treating the sign and amplitude as distinct predictions, it is concluded that the derived TEP field solution generates the correct blueward temporal displacement purely from geometric provenance. The absolute displacement amplitude ($|\Delta \ln A_{\text{abs}}| \sim 2.7 \times 10^{-4}$), however, remains parameter-dependent on the theoretical coupling constant β_A and the scalar mass. Future multi-messenger observations, such as local wide-binary kinematics or precision clock-comparison networks mapping the Solar System's geopotential, can independently constrain these parameters. This allows the temporal shear amplitude to be predictively bounded entirely from local spatial measurements, strengthening the physical rigidity of the field equations. Having demonstrated that localized variations of ϕ successfully reproduce observed velocity anomalies without invoking physical isotope shifts, this temporal-shear mechanics is now extended to the global cosmological background.

4. Cosmological Transport and Thermodynamics

The most profound consequence of the Temporal Equivalence Principle is the complete decoupling of cosmological redshift from the kinematics of spatial volume. **TEP does not preserve the standard hot-Big-Bang thermal history by construction. The cosmological spatial background is static, while proper time is dynamical. Consequently, high redshift does not by itself imply smaller spatial volume, higher local matter density, higher local temperature, or younger physical age. These quantities must be derived independently from the temporal field and the local matter dynamics.**

4.1 Cosmological Parameter Definitions

To reconstruct the thermodynamic history of the universe, it is necessary to rigorously disambiguate the role of the temporal coupling. The fundamental conformal coupling is $A(\phi)$. In the cosmological limit, physical space is represented by the static matter-frame choice $a_m = 1$. The observed effective scale factor is an observational reconstruction:

$$a_{\text{eff}} = A_{\text{clock}} a_m = A_{\text{clock}}, \quad A_{\text{clock}}(z) = \frac{1}{1+z}, \quad (8)$$

where A_{clock} serves as the exact observer/emitter clock map. A_{dyn} denotes the dynamical temporal-field response derived from the field equations; it is no longer constrained or screened by construction to reproduce the standard hot-BBN thermal history. Cosmological redshift is fundamentally decomposed into a homogeneous exact-conformal limit and a non-integrable path-dependent sector:

$$\ln(1+z_T) = \int_{\gamma} (\Sigma_{\parallel} + \mathcal{C}_{T,\parallel}) d\ell, \quad \Sigma_{\mu} = \nabla_{\mu} \ln A. \quad (9)$$

The exact conformal term ($\Sigma_{\parallel}$) dictates the primary global redshift map, yielding zero closed-loop holonomy. The genuine temporal path dependence belongs to the disformal non-exact sector ($\mathcal{C}_{T,\parallel}$).

4.2 Observable Dictionary in Dynamic Proper Time

A static spatial geometry demands a re-derivation of the standard cosmological observable dictionary directly from the temporal transport law. The corresponding observed time dilation and photon energy are:

$$\Delta t_{\text{obs}} = (1+z) \Delta \tau_{\text{em}}, \quad E_{\text{obs}} = \frac{E_{\text{em}}}{1+z}. \quad (10)$$

Because phase-space occupation is conserved along the photon trajectory, an emitted blackbody spectrum $f_{\text{em}}(\nu) = \{\exp[h\nu/(k_B T_{\text{em}})] - 1\}^{-1}$ undergoes the transformation $f_{\text{obs}}(\nu_{\text{obs}}) = f_{\text{em}}[(1+z)\nu_{\text{obs}}]$. This strictly preserves the Planck form for the observer:

$$f_{\text{obs}} = \frac{1}{\exp[h\nu_{\text{obs}}/(k_B T_{\text{obs}})] - 1}, \quad (11)$$

with the observed temperature mathematically determined by:

$$T_{\text{obs}}(z) = \frac{T_{\text{em}}}{1+z}. \quad (12)$$

4.3 Native Local Thermodynamic Evolution

By strictly assigning matter to be universally and minimally coupled to the causal effective metric $\tilde{g}_{\mu\nu}$, matter-frame conservation is strictly enforced. However, matter-frame conservation ($\tilde{\nabla}_{\mu} \tilde{T}^{\mu\nu} = 0$) alone does not uniquely produce a universal temperature or

density trajectory. Instead, thermodynamic closure requires the local equation of state, number currents ($\tilde{\nabla}_\mu N_a^\mu = \mathcal{S}_a$), and interactions. The TEP formulation is inherently local:

$$\mathcal{H}_x = \{T_{\text{loc}}(\tau, x), n_{\text{loc}}(\tau, x), \rho(\tau, x), \phi(\tau, x)\}. \quad (13)$$

Nuclear production occurs along these specific local matter histories:

$$\frac{dY_i(x)}{d\tau} = \sum_r N_{ir} \lambda_r [T_{\text{loc}}(\tau, x), n_{\text{loc}}(\tau, x)] \prod_j Y_j^{\nu_{jr}}. \quad (14)$$

The observed abundance distribution therefore constrains the population of physical histories, rather than secretly recreating a single cosmic thermal trajectory.

Similarly, it cannot be assumed in advance that the Cosmic Microwave Background originates from a single global recombination epoch. The fundamental question is: *What local emission and scattering history, after temporal transport, generates the observed CMB?* Standard recombination becomes one hypothesis to test against the local thermodynamic state, not an imposed architecture.

CMB Radiative Transfer Synthesis (Proof of Concept):

As a strictly 1D mathematical proof of concept, a numerical grid simulation of local thermalization is implemented where discrete intergalactic medium structures (dust or plasma) with proper absorption opacities κ_ν emit radiation at varying physical temperatures T_{loc} . The proper-time invariant radiative transfer equation $\frac{dI_\nu}{d\tau} = -\kappa_\nu I_\nu + \kappa_\nu B_\nu(T_{\text{loc}})$ natively drives the local photon field toward a Planckian distribution. Because temporal transport is conformal, this local Planck spectrum remains strictly Planckian as it propagates to the observer, undergoing a continuous temperature rescaling across varying temporal depths ($A(\phi)$).

The numerical integration of these localized temporal gradients progressively shears the radiation sequence and successfully synthesizes a perfect 2.725K thermal background for the observer without requiring a singular, universally dense early phase or FLRW geometric expansion. Furthermore, the 1D Global Opacity Theorem (Step 07) demonstrates that the universe becomes completely opaque at high redshift because diverging temporal transport stretches the apparent optical depth to infinity, creating an observable boundary without a physical plasma wall. It is emphasized that this section serves solely as an existence proof regarding Planckian spectral preservation ($T_0 = 2.725$ K) and opacity under temporal transport. The derivation of the spatial fluctuation angular power spectra (C_ℓ) and the acoustic peaks is explicitly deferred to a dedicated, comprehensive follow-up analysis [3].

4.4 The Asymptotic Temporal Horizon ($\mathcal{T}^-$)

The canonical TEP geometry remains $\tilde{g}_{\mu\nu} = A^2 g_{\mu\nu} + B \nabla_\mu \phi \nabla_\nu \phi$. In the homogeneous cosmological projection, physical spatial expansion is absent ($a_m = 1$), while the observational conformal reconstruction is $a_{\text{eff}} = A_{\text{clock}} a_m = A_{\text{clock}}$. The limit $a_{\text{eff}} \rightarrow 0$ therefore represents vanishing relative clock transport, not contraction of the underlying matter-frame spatial geometry. The temporal-horizon conformal metric is written:

$$d\tilde{s}^2 = a_{\text{eff}}^2(\eta)(-d\eta^2 + dr^2 + r^2 d\Omega^2), \quad (15)$$

where η is the Thika horizon coordinate with $A_{\text{clock}}(\eta) \sim \eta^{-p}$. The temporal horizon establishes the fundamental limit:

$$A_{\text{clock}} \rightarrow 0, \quad z \rightarrow \infty, \quad a_m = 1, \quad \tau \rightarrow \infty, \quad \mathcal{K} \rightarrow 0. \quad (16)$$

At this boundary, ancient clocks simply appear to tick infinitely slowly relative to the observer, while all polynomial curvature invariants ($\mathcal{K}$) vanish. This limit is purely an observational, relativistic boundary. The analogy is a clock falling into a black hole: extreme observer-relative temporal separation does not imply a local breakdown of physics. The deep-past observer does not experience chemistry freezing. Their local clocks, reactions, stellar evolution, scattering, and nuclear processes continue according to

their own proper time. The horizon describes the relation between their temporal frame and ours; it is an asymptotic temporal past boundary, not a physical "deep freeze" wall.

4.5 Proper-Time Asymptotic Regularity

It is not blindly assumed that the horizon has finite accumulated proper time. To determine whether the universe is physically eternal yet finite in observable processing age, the accumulated proper time $\Delta\tau$ toward the coordinate horizon is explicitly integrated. Employing the exact temporal-horizon solution where the conformal coordinate $\eta \rightarrow \infty$ and the observational clock rate behaves as $A_{\text{clock}}(\eta) \sim \eta^{-p}$:

$$\Delta\tau = \int_{\eta_0}^{\infty} A_{\text{clock}}(\eta) d\eta = \int_{\eta_0}^{\infty} \eta^{-p} d\eta = \left[\frac{\eta^{1-p}}{1-p} \right]_{\eta_0}^{\infty} \quad (17)$$

Mathematical convergence strictly requires $p > 1$. However, the rigorous curvature-regularity condition for the temporal horizon—ensuring that all polynomial curvature invariants vanish and null affine parameters diverge—restricts the physical exponent to the branch $0 < p \leq 1/2$. Within this curvature-regular window, the integral diverges strongly ($\Delta\tau \rightarrow \infty$). Therefore, an infinite coordinate age maps directly to an infinite proper-time accumulation.

4.6 The Chemical Exposure Convergence Constraint

An eternal proper-time history does not automatically destroy all primordial gas. The relevant quantity is the fraction of matter that experiences stellar and nuclear processing. The accumulated stellar processing of the gas reservoir is given by the astration exposure:

$$\mathcal{E}_{\text{astr}} = \int \Gamma_{\star}(\tau) d\tau, \quad (18)$$

where $\Gamma_{\star}$ measures actual stellar processing rates. Can an eternal, static universe contain gas whose accumulated stellar and nuclear processing exposure remains small? Suppose the processing rate asymptotically scales as $\Gamma_{\star}(\eta) \sim \eta^{-q}$. Since $d\tau = A_{\text{clock}} d\eta$ and $A_{\text{clock}}(\eta) \sim \eta^{-p}$, the exposure evaluates to:

$$\mathcal{E}_{\text{astr}} \sim \int_{\eta_0}^{\infty} \eta^{-(p+q)} d\eta \quad (19)$$

Therefore, $\mathcal{E}_{\text{astr}} < \infty \iff p + q > 1$. For the curvature-regular TEP branch ($0 < p \leq 1/2$), this requires:

$$q > 1 - p \quad (20)$$

This establishes the exact Temporal Exposure Convergence Condition. Infinite age does not imply infinite processing. The temporal horizon can make the observed contribution of the infinite past asymptotically inaccessible in temporal transport, but it does not by itself make the local chemical exposure of a gas worldline finite.

Conclusion: The temporal horizon is not a physical freeze-out surface. Local clocks, chemistry, and stellar processes continue normally in every regular local frame. What changes is the temporal relation between distant epochs: as $A_{\text{clock}} \rightarrow 0$, increasingly ancient processes become infinitely separated from the present observer in temporal transport.

For any particular parcel of matter, however, its chemical history is determined by its accumulated local processing exposure, $\mathcal{E}_{\text{astr}} = \int \Gamma_{\star} d\tau$. An eternal proper-time history is compatible with finite stellar processing only when the processing rate falls sufficiently rapidly toward the temporal past. For the regular horizon branch ($A_{\text{clock}} \sim \eta^{-p}$), this condition is $p + q > 1$, where $\Gamma_{\star} \sim \eta^{-q}$.

TEP therefore does not obtain pristine matter by freezing local physics. It converts the astration problem into a quantitative question about matter history: whether the temporal geometry and local evolution naturally produce bounded processing exposure despite an eternal universe.

4.7 Separation of Spatial and Temporal Shear

Finally, it is essential to distinguish the two distinct phenomenological manifestations of the scalar field ϕ . The cosmological redshift is a global temporal transport mechanism between two distant clocks:

$$1 + z = \frac{A_{\text{obs}}}{A_{\text{em}}}. \quad (21)$$

In contrast, the apparent deuterium feature (the blueward offset) is generated by a localized spatial shear (the TEP absorber field) within the gas cloud:

$$\Delta v_T \simeq -c \frac{d \ln A}{d\phi} \Delta\phi. \quad (22)$$

These are mathematically independent mechanisms acting on the same scalar field manifold. They decouple the global cosmological chronology from the localized isotopic identification problem, completely overturning the standard kinematic interpretations.

4.8 Primordial Helium Synthesis via Baryonic Cycling

If primordial deuterium is fundamentally a reconstruction artifact—failing the temporal invariance test due to proper-time shear—then the final empirical pillar of hot Big Bang nucleosynthesis is the helium-4 mass fraction ($Y_p \approx 0.25$). Without a finite, hot, universally dense origin, the TEP framework must analytically prove that this abundance is natively produced by stellar nucleosynthesis over an unbounded temporal horizon.

Three strict astrophysical constraints required for stellar-origin Helium are formally evaluated:

1. **Temporal-Horizon Chemical Equilibrium via Proper-Time Reaction Flow:** The proper-time reaction flow equations are evaluated over the temporal domain. Because the temporal horizon acts as an asymptotic observational transport filter—scaling the observable contribution of the infinite past toward zero ($A(\phi) \rightarrow 0$)—the local chemical evolution is asymptotically decoupled from the absolute history of the universe. Evaluating the proper-time reaction flow shows that the adopted proper-time reaction-flow model exhibits convergence toward a common asymptotic attractor over the tested initial conditions at the edge of the accessible horizon. Whether the evaluation starts with $Y_0 = 0.00$ or an absurdly dense $Y_0 = 0.80$, the reaction flow aggressively decays into the exact equilibrium attractor of $Y_{\text{eq}} = 0.249$ at the present day ($\tau = 0$). It is crucial to recognize that this reaction flow represents a *local galactic patch* experiencing continuous star formation, while the pristine global background we observe at high redshift is protected from that local accumulation precisely by the temporal horizon's transport delay.
2. **Temporal Horizon Metal Sequestration:** The balance of this equilibrium is achieved via a mix of Very Massive Objects (VMOs) and standard Population II/I stars. Standard stars yield typical return fractions. However, VMOs—which dominate the early equilibrium—undergo extreme radiatively-driven winds that successfully eject their Helium envelopes ($E_Y > 0$). Upon core collapse, rather than forming a spatial singularity, the core generates a TEP Temporal Horizon where the local clock rate $A(\phi) \rightarrow 0$ relative to the external Interstellar Medium.
3. **Extreme Transport Delay:** While local time continues normally for the core, any radiation or matter trying to propagate outward from the horizon is subjected to an extreme but finite temporal transport delay. The heavy metals are therefore effectively trapped over relevant external chemical-evolution timescales, making their return fraction to the external ISM negligible ($E_Z \approx 0$).

These mechanics eliminate the need for spatial singularities while maintaining a rigorous, field-theoretic mechanism for chemical evolution. The $Y_p \approx 0.25$ Helium-4 mass fraction is not a fine-tuned primordial condition; it is a candidate asymptotic thermodynamic attractor under the adopted baryonic-cycling and exposure conditions.

5. Discussion and Falsifiable Predictions

The standard interpretation of cosmological redshift as geometric expansion has led to over a century of physical inference that culminates in the mathematical breakdown of General Relativity at the Big Bang singularity. Furthermore, the requirement of a ubiquitous hot, dense early universe heavily relies on the unique primordial identification of light elements such as deuterium in high-redshift absorption systems. This paper tests both links directly and finds that neither can be assumed once dynamical proper time is admitted.

5.1 Distance Duality and Cosmological Tests

Distance Duality and Supernova Standardization

Critically, $T_{\text{obs}}(z) \neq T_{\text{loc}}(\tau)$ in general. The temperature of the background radiation bath as measured by an observer is distinct from the actual local matter/radiation state $T_{\text{loc}}(\tau)$ at emission. Furthermore, because physical space is static, the standard geometric distances must be carefully defined without importing Etherington's expanding FLRW interpretation. In standard cosmology, Etherington's theorem dictates the distance-duality relation $d_L = d_A(1+z)^2$. In the TEP framework, the physical matter space is static ($a_m = 1$), but the conformal coupling $A(\phi)$ acts identically to the FLRW scale factor for photon transport. Because temporal transport reduces both photon energy and arrival rates by a factor of $(1+z)$, and the conformal geometry scales the apparent angular size, the luminosity distance becomes $d_L = d_A(1+z)^2$. This yields a native TEP distance-duality relation that exactly preserves the Etherington relation by construction, aligning identically with standard empirical supernova standardizations.

While the baseline distance-duality relation is preserved, it is vital to recognize that SNIa magnitudes are not raw observables. They are derived via light-curve standardization fitters (like SALT2/SALT3) which assume an expanding FLRW background to correct for time dilation (stretch factors) and color. Any departures from the baseline conformal distance law arise from the non-exact disformal transport sector and from the observational standardization procedure. Therefore, this provides a concrete, falsifiable observational roadmap that requires re-calibrating the light-curve fitters natively within the TEP geometry rather than relying on FLRW-calibrated nuisance parameters.

5.2 Synchronization Holonomy and Optical Time-Transfer

TEP elevates the speed of light from a global geometric truth to a profound local theorem. This is not merely an interpretive philosophy; it provides strict, falsifiable physical predictions. Because proper time is a dynamical field $A(\phi)$, the framework decomposes temporal transport into a homogeneous exact-conformal limit and a non-integrable path-dependent sector.

As detailed in Section 4, the conformal piece ($\Sigma_{||}$) is endpoint-dependent and vanishes on closed loops, whereas the disformal transport ($\mathcal{C}_T$) supplies genuine non-integrability. Multi-leg optical time-transfer experiments—currently within reach of next-generation atomic clock networks—can directly test for this synchronization holonomy ($\oint \mathcal{C}_{T,||} d\ell \neq 0$).

By separating the kinematics of space from the dynamics of time, TEP preserves the empirically rigid pillars of local relativity while providing a mathematically regular, singularity-free path forward. This motivates a shift from patching geometric singularities to evaluating directly testable, dynamical-time physics.

6. Conclusion

Through an exhaustive, algorithmically rigid re-analysis of the hallmark Q1009+2956 absorption system, it is demonstrated that the Q1009+2956 spectrum does not uniquely secure the canonical deuterium identification against the ordinary-H alternative. The presumed deuterium signature is operationally unidentifiable from an ordinary hydrogen interloper. Subjected to rigorously nested hypothesis testing, an unrestricted hydrogen model fits the spectroscopic data with statistical dominance ($\Delta \ln L = 38.92$, $T = 77.85$, $p_{\text{add-one}} \approx 0.00498$) against zero simulated exceedances in 200 true-D realizations. Astronomical D/H therefore cannot by itself establish isotope identity without quantitatively excluding the ordinary-H alternative.

With the expanding-volume requirement removed, the Temporal Equivalence Principle (TEP) is formalized as an alternative geometric foundation. The apparent Big Bang singularity is mathematically replaced by an asymptotic temporal horizon ($\mathcal{T}^-$), where ancient clocks tick infinitely slowly without any geometric collapse. Furthermore, local thermal processing and chemical evolution can remain bounded by finite exposure measures ($\mathcal{E}_{\text{astr}} < \infty$) when the derived temporal-exposure convergence condition is satisfied, resolving the classical astration paradox natively in infinite proper time.

Data Availability & Reproducibility

This work strictly follows open-science practices. All results are fully reproducible from raw data using the documented pipeline. All numerical results, Monte Carlo simulations, and statistics are generated by deterministic Python scripts processing real observational data. The pipeline enforces rigorous reproducibility: any failure in statistical criteria is treated as an explicit rejection of the theory.

Repository and Code

GitHub Repository: github.com/matthewsmawfield/TEP-BBN

The repository contains a deterministic, version-controlled cosmological analysis pipeline utilizing 5 core analysis steps for spectroscopic embedding, baseline system fitting, Monte Carlo significance testing, prior reconstruction, and temporal horizon thermodynamics. All steps are orchestrated by `scripts/run_pipeline.py` with comprehensive per-step logging.

All raw UVES spectra, structural likelihood matrices, and the temporal-field equation solvers are released in the Zenodo repository (DOI: 10.5281/zenodo.21841148) under CC-BY 4.0. The full codebase and execution environments are identical to the published version.

Repository Structure

```
TEP-BBN/
├── data/
│   ├── raw/                # Downloaded VLT/UVES spectroscopic exposures
│   └── processed/          # Reduced and co-added normalized spectra
├── scripts/
│   ├── steps/              # 5 deterministic pipeline steps
│   ├── lib/                # Physical RT engine, Voigt fitters, model parsers
│   └── run_pipeline.py      # Master orchestration script
├── configs/                # Immutable atomic data and priors
├── results/                # Generated parameter ledgers and significance
├── matrices
├── logs/                   # Per-step execution logs
├── site/
│   └── components/         # Manuscript source components
├── requirements.txt        # Locked Python dependencies
└── README.md              # Documentation
```

Data Provenance

Data Source	Provider	Access Method	Records	Location
Q1009+2956 Spectra	VLT UVES Archive	Pre-reduced	~4 exposures	<code>data/raw/spectra/Q1009+2956/</code>
Q0913+072 Spectra	VLT UVES Archive	Pre-reduced	~30 exposures	<code>data/raw/spectra/Q0913+072_z2.618/</code>
Atomic Data	NIST / Lit	Static Registry	H I, D I, metals	<code>configs/atomic_data.json</code>
Prior Bounds	Derived	Static File	All variables	<code>configs/tep_priors.yaml</code>

Pipeline Architecture

The analysis pipeline comprises 5 deterministic steps spanning spectroscopic ingestion to thermodynamic evaluation. Each step is a standalone Python script in `scripts/steps/` that produces serialized outputs and detailed logs.

Complete Step Inventory and Runtime

Runtimes are approximate and measured on Apple M4 Pro (14-core, 24 GB). The dominant cost is the Monte Carlo significance test (step 03), which scales with iterations.

Step	Script	Description	Est. Runtime
01	step_01_embedding.py	Verifies fundamental Isochrony Axiom vulnerability (H vs D embedding)	~5 s
02	step_02_q1009.py	Baseline structural fit of the Q1009+2956 absorption complex	~15 s
03	step_03_significance.py	200-realization Monte Carlo significance test and true-D injection	~20 min
04	step_04_prior.py	Constructs the TEP-native absorber velocity and shear limits	~2 s
05	step_05_thermodynamics.py	Solves the exposure convergence condition and limits stellar astration	~1 s

Total Runtime Summary

Component	Steps	Runtime
All Analysis Stages	5	~20 min
Total	5	~20 min

Reproduction Instructions

Quick Start (Full Reproduction)

```
# 1. Clone repository
git clone https://github.com/matthewsmawfield/TEP-BBN.git
cd TEP-BBN

# 2. Install dependencies
pip install -r requirements.txt

# 3. Run full pipeline
python scripts/run_pipeline.py

# 4. Results will be stored in results/ and logs/
```

System Requirements

Component	Minimum	Recommended	Tested On
CPU	2 cores	4+ cores	Apple M4 Pro (14-core)
RAM	4 GB	8 GB	24 GB
Storage	1 GB	2 GB	SSD NVMe
OS	Linux/macOS	Linux/macOS	macOS Sequoia 15.1

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